

Sheaf Theory

Labix

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Abstract

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1 Introduction to Presheaves and Sheaves

1.1 The Category of Presheaves of Sets

Definition 1.1.1 (The Category of Open Sets)

Let $(X, \mathcal{T})$ be a topological space. Define the category $\mathbf{Open}(X)$ to consist of the following data.

- The objects are given by the open sets $\mathcal{T}$ of X .
- For two open sets $U, V \in \mathcal{T}$, define the morphisms from U to V by

$$\mathrm{Hom}_{\mathbf{Open}(X)}(U, V) = \begin{cases} \{\iota : U \hookrightarrow V\} & \text{if } U \subseteq V \\ \emptyset & \text{otherwise} \end{cases}$$

- Composition is given by the composition of functions.

Definition 1.1.2 (Presheaves on a Space)

Let X be a space. A presheaf on X is a contravariant functor $\mathcal{F} : \mathbf{Open}(X) \rightarrow \mathbf{Set}$.

Example 1.1.3 (The Constant Presheaf)

Let X be a space. Let S be a set. Define the constant presheaf with value S to be the presheaf $\underline{S}_{\mathrm{pre}}$ defined as follows.

- For each open set $U \subseteq X$, we have

$$\underline{S}_{\mathrm{pre}}(U) = S$$

- For each inclusion $V \subseteq U$ of open sets, define

$$\mathrm{res}_V^U = \mathrm{id}_S$$

Definition 1.1.4 (The Category of Presheaves)

Let X be a topological space. Define the category of presheaves on X to be

$$\mathbf{PSh}(X) = \mathrm{Func}(\mathbf{Open}(X)^{\mathrm{op}}, \mathbf{Set})$$

Proposition 1.1.5

Let X be a space. Then $\mathbf{PSh}(X)$ admits all finite limits.

We now have the definition of a morphism of presheaves, which is just a natural transformation, and the definition of isomorphisms, which is just a natural isomorphism.

Definition 1.1.6 (The Stalk of a Presheaf)

Let X be a space. Let $\mathcal{F}$ be a presheaf on X . Let $p \in X$. Consider the full subcategory $\mathcal{J}_p$ of $\mathbf{Open}(X)$ consisting of open sets in X that contain p . Define the stalk of $\mathcal{F}$ at p to be the colimit

$$\mathcal{F}_p = \mathrm{colim}_{\mathcal{J}_p} \mathcal{F}(V_p)$$

Definition 1.1.7 (Induced Map of Stalks)

Let X be a topological space. Let $\mathcal{F}, \mathcal{G}$ be presheaves on X . Let $p \in X$. Define the induced map of stalks

$$\phi_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$$

to be the map induced by the universal product of colimits.

1.2 The Category of Sheaves of Sets

Definition 1.2.1 (Sheaves)

Let X be a space. A sheaf on X is a presheaf $\mathcal{F} : \mathbf{Open} \rightarrow \mathbf{Sets}$ such that the following are true.

- **Identity:** If $\{U_i | i \in I\}$ is an open cover of U and $\phi_1, \phi_2 \in \mathcal{F}(U)$ and $\phi_1|_{U_i} = \phi_2|_{U_i}$ for all i , then $\phi_1 = \phi_2$
- **Gluing:** If $\{U_i | i \in I\}$ is an open cover of U and $\phi_i \in \mathcal{F}(U_i)$ for all $i \in I$ such that $\phi_i|_{U_i \cap U_j} = \phi_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there exists some $\phi \in \mathcal{F}(U)$ such that $\phi|_{U_i} = \phi_i$ for all $i \in I$.

Example 1.2.2

Let X, Y be spaces. Define a presheaf $\mathcal{F}$ on X by

$$\mathcal{F}(U) = \{f : U \rightarrow Y \mid f \text{ is continuous}\}$$

for $U \subseteq X$ an open set, together with restriction maps. Then $\mathcal{F}$ is a sheaf.

Proof

Identity: Suppose that $f, g \in \mathcal{F}(U)$ for some open set $U \subseteq X$ and $f|_{U_i} = g|_{U_i}$ for $\{U_i \mid i \in I\}$ an open cover of U . Then for any $x \in U$, there exists $i \in I$ such that $x \in U_i$, and so

$$f(x) = f|_{U_i}(x) = g|_{U_i}(x) = g(x)$$

Hence $f = g$.

Gluing: Suppose that $f_i \in \mathcal{F}(U_i)$ for $\{U_i \mid i \in I\}$ an open cover of an open set $U \subseteq X$ such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$. Define $f \in \mathcal{F}(U)$ by $f(x) = f_i(x)$ for $x \in U_i$. This is well defined since $\{f_i\}$ agree on intersections. It remains to show that f is continuous. Let $W \subseteq Y$ be an open set. I claim that $f^{-1}(W) = \bigcup_{i \in I} f_i^{-1}(W)$. Let $x \in f^{-1}(W)$. Then $f(x) \in W$. Since $x \in W \subseteq U = \bigcup_{i \in I} U_i$, there exists $i \in I$ such that $x \in U_i$. Hence $f_i(x) = f(x) \in W$. Thus $x \in f_i^{-1}(W)$. Conversely, suppose that $x \in f_i^{-1}(W)$. Then clearly $f(x) = f_i(x) \in W$ so that $x \in f^{-1}(W)$. Thus $f^{-1}(W) = \bigcup_{i \in I} f_i^{-1}(W)$. Since each f_i is continuous and W is an open set, $f_i^{-1}(W)$ is open and so the arbitrary union of $f_i^{-1}(W)$ is open. Hence $f^{-1}(W)$ is open. ■

Example 1.2.3 (Locally Constant Sheaf)

Let X be a space. Let S be a set. Define the locally constant presheaf $\underline{S}$ of X with values in S as follows.

- For each open set $U \subseteq X$, define

$$\underline{S}(U) = \{f : U \rightarrow S \mid f \text{ is continuous where } S \text{ has the discrete topology}\}$$

- For each inclusion $V \subseteq U$ of open subsets, define the induced map $\underline{S}(U) \rightarrow \underline{S}(V)$ by $f \mapsto f|_V$.

The following are true regarding the locally constant sheaf.

- $\underline{S}$ is a sheaf.
- Let $U \subseteq X$ be an open set. Let $\mathcal{B}$ be the set of connected components of U . Then

$$\underline{S}(U) \cong \prod_{C \in \mathcal{B}} S$$

- For all $p \in X$, $\underline{S}_p = S$.

Proof

Let Y be a connected component of U . Let $f \in \underline{S}(Y)$. I claim that f is constant. Let $y_0 \in Y$. Define $A = \{y \in Y \mid f(y) = f(y_0)\} = f^{-1}(f(y_0))$ and $B = \{y \in Y \mid f(y) \neq f(y_0)\}$. Clearly $Y = A \cup B$ and $A \cap B = \emptyset$. Both A and B are open sets since S has a discrete topology. Since Y is connected and A is non-empty, this implies that $B = \emptyset$. Hence $Y = A$ and f is constant on Y . Thus there is

a bijection $\underline{S}(Y) \cong S$.

Define a function $\underline{S}(U) \rightarrow \prod_{C \in \mathcal{B}} S$ as follows. For each connected component $C \in \mathcal{B}$ of U , choose $x_C \in C$. Then the map is defined by $f \mapsto (f(x_C))$. This map is injective since $f(x_C) = g(x_C)$ implies that $f(C) = g(C)$ for all connected components $C \in \mathcal{B}$. Hence $f = g$. It is surjective since for any choice of sequence $(x_C)_{C \in \mathcal{B}}$, the map $f : U \rightarrow S$ defined by $f(x) = x_C$ for $x \in C$ is continuous because S is discrete and $\{x_C\}$ is a closed set, and $f^{-1}(x_C) = C$ is a connected component and so is a closed set. ■

Proposition 1.2.4

Let X be a space. Let $\mathcal{F}$ be a presheaf. Then the following are equivalent characterizations.

- $\mathcal{F}$ is a sheaf
- For any open covering $U = \bigcup_{i \in I} U_i$, the sequence

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{f} \prod_{i \in I} \mathcal{F}(U_i) \xrightarrow{g} \prod_{i,j \in I} \mathcal{F}(U_i \cap U_j)$$

where f is given by the product of the unique maps $\mathcal{F}(U) \rightarrow \mathcal{F}(U_i)$ and g is given by sending $(s_i)_{i \in I}$ to $(s_i|_{U_i \cap U_j} - s_j|_{U_i \cap U_j})_{i,j \in I}$ is exact.

- For any open covering $U = \bigcup_{i \in I} U_i$, the following diagram

$$\mathcal{F}(U) \xrightarrow{f} \prod_{i \in I} \mathcal{F}(U_i) \xrightleftharpoons[h]{g} \prod_{i,j \in I} \mathcal{F}(U_i \cap U_j)$$

is an equalizer diagram, where f is given by the product of the unique maps $\mathcal{F}(U) \rightarrow \mathcal{F}(U_i)$, g is given by sending $(s_i)_{i \in I}$ to $(s_i|_{U_i \cap U_j})_{i,j \in I}$, h is given by sending $(s_i)_{i \in I}$ to $(s_j|_{U_i \cap U_j})_{i,j \in I}$.

Proof

We first note that the second and third condition are equivalent. Therefore we just have to show that the first condition is equivalent to the third one. Let $\mathcal{F}$ be an abelian sheaf. We want to show that $\text{im}(f) = \{x \in \prod_{i \in I} \mathcal{F}(U_i) \mid g(x) = h(x)\}$. Therefore we begin with $x \in \text{im}(f)$. Then $f(x) = (x_i)_{i \in I}$. But we have that $g((x_i)_{i \in I}) = (x_i|_{U_i \cap U_j})_{i,j \in I}$ and $h((x_i)_{i \in I}) = (x_j|_{U_i \cap U_j})_{i,j \in I}$. We want to show that for all $i, j \in I$, we have that $x_i|_{U_i \cap U_j} = x_j|_{U_i \cap U_j}$. But x_i is just the restriction of x to U_i and x_j is the restriction of x to U_j . But by definition of a sheaf, restricting to V_i and further to V_{ij} is the same as directly restricting to V_{ij} . Similarly, restricting to V_j and further to V_{ij} is the same as directly restricting to V_{ij} . We conclude that applying g and h to $(x_i)_{i \in I}$ give the same element. Thus $x \in \{x \in \prod_{i \in I} \mathcal{F}(U_i) \mid g(x) = h(x)\}$.

Now let $(y_i)_{i \in I} \in \{x \in \prod_{i \in I} \mathcal{F}(U_i) \mid g(x) = h(x)\}$. This means that $y_i|_{U_i \cap U_j} = y_j|_{U_i \cap U_j}$ for all $i, j \in I$. By the gluing axiom, there exists $y \in \mathcal{F}(U)$ such that $y|_{U_i} = y_i$. Thus y is in the image of f and we conclude that $y \in \text{im}(f)$.

Finally, we also need to show that f is injective. Suppose that $x, y \in \mathcal{F}(U)$ is such that $x_i = y_i$ for all $i \in I$. Then by the identity axiom we conclude that $x = y$. Thus f is injective. Thus the equalizer condition is satisfied.

Now suppose that the equalizer condition is satisfied. Then f is injective. This means that for $x, y \in \mathcal{F}(U)$ such that $x_i = y_i$ for all $i \in I$, then $x = y$. But this is precisely the identity axiom. Suppose that $\text{im}(f) = \text{Eq}(g, h)$. Let $s_i \in \mathcal{F}(U_i)$ such that

$$s_i|_{U_i \cap U_j} = g(s_i) = h(s_i) = s_j|_{U_i \cap U_j}$$

for all $i, j \in I$. Then $(s_i)_{i \in I} \in \text{Eq}(g, h) = \text{im}(f)$. Which means that there exists some $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$. But this is the content of the gluing axiom. Since the identity and the gluing axiom are satisfied, we conclude that $\mathcal{F}$ is a sheaf. ■

Proposition 1.2.5 (Gluing Sheaves)

Let X be a space. Let $\{U_i \mid i \in I\}$ be an open cover of X . Let $\mathcal{F}_i$ be a sheaf of sets on U_i . Let $\phi_{i,j} : \mathcal{F}_i|_{U_i \cap U_j} \rightarrow \mathcal{F}_j|_{U_i \cap U_j}$ be a collection of morphism of sheaves for ranging $i, j \in I$ such that

the following are true.

- For each $i, j \in I$, $\phi_{i,j}$ is an isomorphism.
- For each $i, j, k \in I$, we have $\phi_{j,k} \circ \phi_{i,j} = \phi_{i,k}$.

Then there exists a unique (up to unique isomorphism) sheaf of sets $\mathcal{F}$ on X such that there are isomorphisms $\mathcal{F}|_{U_i} \xrightarrow{\cong} \mathcal{F}_i$ for all $i \in I$.

Definition 1.2.6 (The Category of Sheaves of Sets)

Let X be a space. Define the category of sheaves of sets $\mathbf{Sh}(X)$ on X to be the full subcategory of $\mathbf{PSh}(X, \mathbf{Set})$ consisting of sheaves of sets.

Proposition 1.2.7

Let X be a space. Then $\mathbf{Sh}(X)$ admits all finite limits.

Proposition 1.2.8

Let X be a space. Let $\mathcal{F}, \mathcal{G}$ be sheaves on X . Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism. Then φ is an isomorphism if and only if $\varphi_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$ is a bijection for all $p \in X$.

Proposition 1.2.9

Let X be a space. Let $\mathcal{F}, \mathcal{G}$ be sheaves on X . Let $\varphi, \psi : \mathcal{F} \rightarrow \mathcal{G}$ be morphisms. Then $\varphi = \psi$ if and only if $\varphi_p = \psi_p$ for all $p \in X$.

1.3 The Sheafification Functor

Definition 1.3.1 (Compatible Germs)

Let X be a space. Let $\mathcal{F}$ be a presheaf on X . Let $[(s_p, U_p)]_{p \in U} \in \prod_{p \in U} \mathcal{F}_p$. We say that the element is a compatible germ if for all $p \in U$, there exists an open subset $V_p \subseteq U$ such that

$$[(s_p|_{V_p}, V_p)] = [(s_q, U_q)] \in \mathcal{F}_q$$

for all $q \in V_p$.

Lemma 1.3.2

Let X be a space. Let $\mathcal{F}$ be a presheaf on X . Let $[(s_p, U_p)]_{p \in U} \in \prod_{p \in U} \mathcal{F}_p$. Then the element is a compatible germ if and only if there exists an open cover $\bigcup_{i \in I} U_i$ of U and $f_i \in \mathcal{F}(U_i)$ such that

$$[(f_i, U_i)] = [(s_p, U_p)] \in \mathcal{F}_p$$

for all i and all $p \in U_i$.

Definition 1.3.3 (Sheafification)

Let X be a space. Let $\mathcal{F}$ be a presheaf on X . Define the sheafification of $\mathcal{F}$ to be the sheaf

$$\mathcal{F}^{\text{sh}} : \mathbf{Open}(X) \rightarrow \mathbf{Set}$$

given as follows.

- For each open set $U \subseteq X$, we have

$$\mathcal{F}^{\text{sh}}(U) = \left\{ [(s_p, U_p)]_{p \in U} \in \prod_{p \in U} \mathcal{F}_p \mid [(s_p, U_p)]_{p \in U} \text{ is a compatible germ} \right\}$$

- For each $V \subseteq U$ an inclusion, there is a unique morphism $\mathcal{F}^{\text{sh}}(U) \rightarrow \mathcal{F}^{\text{sh}}(V)$ given by

$$([(s_p, U_p)]_{p \in U} \in \mathcal{F}^{\text{sh}}(U) \mapsto [(s_p, U_p)]_{p \in V}$$

Lemma 1.3.4

Let X be a space. Let $\mathcal{F}$ be a presheaf on X . Then $\mathcal{F}^{\text{sh}}(X)$ is a sheaf.

Definition 1.3.5 (The Universal Morphism)

Let X be a space. Let $\mathcal{F}$ be a presheaf on X . Define the universal morphism $\theta_{\mathcal{F}} : \mathcal{F} \rightarrow \mathcal{F}^{\text{sh}}$ as follows. For each open set $U \subseteq X$, define

$$\theta_{\mathcal{F}}(U) : \mathcal{F}(U) \rightarrow \prod_{p \in U} \mathcal{F}_p$$

by $s \mapsto ((s, U))_{p \in U}$.

Lemma 1.3.6

Let X be a space. Let $\mathcal{F}$ be a presheaf on X . Let $U \subseteq X$ be an open set. Then the following are true.

- $\text{im}(\theta_{\mathcal{F}}(U)) = \mathcal{F}^{\text{sh}}(U)$.
- If $\mathcal{F}$ is a sheaf, then $\theta_{\mathcal{F}}(U) : \mathcal{F} \rightarrow \mathcal{F}^{\text{sh}}$ is injective.

Lemma 1.3.7

Let X be a space. Let $\mathcal{F}$ be a sheaf on X . Then the following are true.

- θ induces an isomorphism

$$\theta_p : \mathcal{F}_p \xrightarrow{\cong} \mathcal{F}_p^{\text{sh}}$$

for all $p \in X$.

- $\theta : \mathcal{F} \rightarrow \mathcal{F}^{\text{sh}}$ is an isomorphism.

Proposition 1.3.8 (The Universal Property of Sheafification)

Let X be a space. Let $\mathcal{F}$ be a presheaf on X . Then the sheafification $\mathcal{F}^{\text{sh}}$ together with $\theta_{\mathcal{F}}$ satisfies the following universal property. For any sheaf $\mathcal{G}$ on X and any morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$, there exists a unique morphism $\varphi^{\text{sh}} : \mathcal{F}^{\text{sh}} \rightarrow \mathcal{G}$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\theta_{\mathcal{F}}} & \mathcal{F}^{\text{sh}} \\ & \searrow \varphi & \downarrow \exists! \varphi^{\text{sh}} \\ & & \mathcal{G} \end{array}$$

Moreover, $\mathcal{F}^{\text{sh}}$ is the unique (up to unique isomorphism) sheaf of X satisfying the above property.

Definition 1.3.9 (The Sheafification Functor)

Let X be a space. Define the sheafification functor

$$(-)^{\text{sh}} : \mathbf{PSh}(X) \rightarrow \mathbf{Sh}(X)$$

as follows.

- Each presheaf $\mathcal{F}$ is sent to its sheafification $\mathcal{F}^{\text{sh}}$.
- For each morphism of presheaves $\mathcal{F} \rightarrow \mathcal{G}$, the morphism of sheaves is given by the universal property of $\mathcal{F}^{\text{sh}}$ applied to the composite $\mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{G}^{\text{sh}}$.

Proposition 1.3.10

Let X be a space. Let $\iota : \mathbf{Sh}(X) \hookrightarrow \mathbf{PSh}(X)$ be the inclusion functor. Then there is an adjunction

$$(-)^{\text{sh}} : \mathbf{PSh}(X) \xrightleftharpoons{+} \mathbf{Sh}(X) : \iota$$

1.4 Pushforward and Pullback Sheaves

Definition 1.4.1 (Direct Image Sheaf)

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Let $\mathcal{F}$ be a presheaf on X . Define the direct image of sheaf

$$f_*\mathcal{F} : \mathbf{Open}(Y) \rightarrow \mathcal{C}$$

as follows.

- For each open set $U \subseteq Y$, define $f_*\mathcal{F}(U) = \mathcal{F}(f^{-1}(U))$.
- For each inclusion $V \subseteq U$, define the induced map

$$\mathcal{F}(f^{-1}(U)) \rightarrow \mathcal{F}(f^{-1}(V))$$

to be the map $\mathcal{F}(\text{incl.} : f^{-1}(V) \hookrightarrow f^{-1}(U))$.

Proposition 1.4.2

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Let $\mathcal{F}$ be a sheaf. Then $f_*\mathcal{F}$ is a sheaf on Y .

Proof

It is clear that $f_*\mathcal{F}$ is a presheaf on Y . We need to check the identity and gluing axiom. By 1.3.4, without loss of generality we can assume that $\mathcal{C} = \mathbf{Set}$.

Identity: Let $U \subseteq Y$ be an open set. Let $\{U_i \mid i \in I\}$ be an open cover of U . Denote the inclusion maps by $\iota_i : U_i \rightarrow U$. Let $s, t \in f_*\mathcal{F}(U)$. Suppose that $f_*\mathcal{F}(\iota_i)(s) = f_*\mathcal{F}(\iota_i)(t)$ for all $i \in I$. This means that $s, t \in \mathcal{F}(f^{-1}(U))$ is such that the images of s, t in $\mathcal{F}(f^{-1}(U_i))$ are equal for all $i \in I$. Since $\{f^{-1}(U_i)\}$ is an open cover of $f^{-1}(U)$ and $\mathcal{F}$ is a sheaf on X , this means that $s = t$.

Gluing: Let $U \subseteq Y$ be an open set. Let $\{U_i \mid i \in I\}$ be an open cover of U . Suppose that $s_i \in f_*\mathcal{F}(U_i) = \mathcal{F}(f^{-1}(U_i))$ is such that the images of s_i and s_j in $f_*\mathcal{F}(U_i \cap U_j) = \mathcal{F}(f^{-1}(U_i \cap U_j))$ agree for all $i, j \in I$. Since $\{f^{-1}(U_i)\}$ is an open cover of $f^{-1}(U)$ and $\mathcal{F}$ is a sheaf on X , we conclude that there exists $s \in \mathcal{F}(f^{-1}(U)) = f_*\mathcal{F}(U)$ such that the image of s in $\mathcal{F}(f^{-1}(U_i)) = f_*\mathcal{F}(U_i)$ is s_i for all $i \in I$. Thus the gluing axiom is satisfied. ■

Example 1.4.3 (Skyscraper Sheaf)

Let X be a space. Let $p \in X$. Let S be a set. Let $\underline{S}$ be the constant sheaf on $\{p\}$. Define the skyscraper sheaf of X at p to be the pushforward

$$\iota_*\underline{S}(U) = \underline{S}(\iota^{-1}(U))$$

where $\iota : \{p\} \hookrightarrow X$ is the inclusion map.

Definition 1.4.4 (The Direct Image Sheaf Functor)

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Let $\mathcal{F}$ be a sheaf on X . Define the direct image sheaf functor

$$f_* : \mathbf{Sh}(X) \rightarrow \mathbf{Sh}(Y)$$

by the following data.

- For each sheaf $\mathcal{F}$ of X , $f_*\mathcal{F}$ is the direct image sheaf on Y .
- For each morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$, $f_*\varphi : f_*\mathcal{F} \rightarrow f_*\mathcal{G}$ is the morphism given by

$$\varphi(f^{-1}(U)) : \mathcal{F}(f^{-1}(U)) \rightarrow \mathcal{G}(f^{-1}(U))$$

Definition 1.4.5 (Inverse Image Sheaf)

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Let $\mathcal{G}$ be a sheaf on Y . Define the inverse image sheaf $f^{-1}\mathcal{G}$ to be the sheafification of the presheaf on X defined as follows.

- For each open set $U \subseteq X$, let $\mathcal{J}(U) \subseteq \mathbf{Open}(X)$ be the full subcategory consisting of open

sets of Y that contain $f(U)$. Then the presheaf sends U to $\lim_{W \in \mathcal{J}(U)} \mathcal{G}(W)$.

- For each inclusion $V \subseteq U$, define the induced map to be induced by the universal property of the limit.

Proposition 1.4.6

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Let $\mathcal{G}$ be a sheaf on Y . Then the inverse image sheaf $f^{-1}\mathcal{G}$ is a sheaf.

Definition 1.4.7 (The Inverse Image Sheaf Functor)

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Let $\mathcal{F}$ be a sheaf on X . Define the inverse image sheaf functor

$$f^{-1} : \mathbf{Sh}(Y) \rightarrow \mathbf{Sh}(X)$$

by the following data.

- For each sheaf $\mathcal{F}$ of Y , $f^{-1}\mathcal{F}$ is the inverse image sheaf on X .
- For each morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$, the morphism $f^{-1}\mathcal{F} \rightarrow f^{-1}\mathcal{G}$ is induced by the universal property of limits.

Example 1.4.8

Let Y be a space. Let $p \in Y$. Let $\iota : \{p\} \hookrightarrow Y$ be the inclusion map. Let $\mathcal{G}$ be a sheaf on Y . Then we have

$$(\iota^{-1}\mathcal{G})(\{p\}) = \mathcal{G}_p$$

Proposition 1.4.9

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Then there is an adjunction

$$f^{-1} : \mathbf{Sh}(Y) \overset{-1}{\rightleftarrows} \mathbf{Sh}(X) : f_*$$

Corollary 1.4.10

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Let $\mathcal{G}$ be a sheaf of Y . Then there is a natural isomorphism

$$(f^{-1}\mathcal{G})_p \cong \mathcal{G}_{f(p)}$$

for any $p \in X$.

Proof f^{-1} is a left adjoint and stalks are colimits. Since left adjoints preserve colimits we are done. ■

Example 1.4.11

Let X be a space. Let $U \subseteq X$ be an open subset. Let $\mathcal{F}$ be a sheaf of sets on X . Let $p \in U$. Then we have

$$(\mathcal{F}|_U)_p \cong \mathcal{F}_p$$

Proof Follows from the above corollary. ■

Lemma 1.4.12

Let Y be a space. Let $U \subseteq Y$ be an open set. Let $\iota : U \hookrightarrow Y$ be the inclusion map. Let $\mathcal{G}$ be a sheaf on Y . Then there is a natural isomorphism

$$\iota^{-1}\mathcal{G} \cong \mathcal{G}|_U$$

Proof Write $\iota^{-1}\mathcal{G}_{\text{pre}}$ the inverse image sheaf before sheafification. Recall that before sheafification, for $W \subseteq U$ an open set we have $(\iota^{-1}\mathcal{G}_{\text{pre}})(W) = \text{colim}_{V \supseteq W} \mathcal{G}(V)$. Since W is the initial object in the category of open sets containing W , we have that the colimit is equal to the terminal object. Hence $(\iota^{-1}\mathcal{G}_{\text{pre}})(W) = \mathcal{G}(W)$. Since $\mathcal{G}$ is already a sheaf, sheafification gives a natural

isomorphism $\iota^{-1}\mathcal{G}) \cong \iota^{-1}\mathcal{G}_{\text{pre}}$ and the latter is equal to $\mathcal{G}|_U$. ■

1.5 Ring, Module and Algebra Objects in Presheaves and Sheaves

Definition 1.5.1 (The Product Sheaf)

Let X be a space. Let $\mathcal{F}, \mathcal{G}$ be presheaves on X . Define the product of $\mathcal{F} \times \mathcal{G}$ as follows.

- For each open set $U \subseteq X$, define $(\mathcal{F} \times \mathcal{G})(U) = \mathcal{F}(U) \times \mathcal{G}(U)$.
- For each inclusion of open sets $V \subseteq U$, the morphism is given by $\text{res}_V^U \times \text{res}_V^U : (\mathcal{F} \times \mathcal{G})(U) \rightarrow (\mathcal{F} \times \mathcal{G})(V)$.

Proposition 1.5.2

Let X be a space. Let $\mathcal{F}, \mathcal{G}$ be presheaves on X . Then the following are true.

- $\mathcal{F} \times \mathcal{G}$ is the categorical product in $\mathbf{PSh}(X)$.
- If $\mathcal{F}$ and $\mathcal{G}$ are sheaves, then $\mathcal{F} \times \mathcal{G}$ is the categorical product in $\mathbf{Sh}(X)$.

Definition 1.5.3 (The Category of Presheaves and Sheaves of Rings)

Let X be space.

- Define the category of presheaves of rings on X to be the category

$$\mathbf{PRing}(X) = \mathbf{Ring}(\mathbf{PSh}(X, \mathbf{Set}), \times, *)$$

of ring objects.

- Define the category of sheaves of rings on X to be the category

$$\mathbf{Ring}(X) = \mathbf{Ring}(\mathbf{Sh}(X), \times, *)$$

of ring objects.

Proposition 1.5.4

Let X be a space. Then the initial object of $\mathbf{Ring}(X)$ is $\mathbb{Z}$.

Definition 1.5.5 (The Category of Presheaves and Sheaves of Modules)

Let X be a space. Let $\mathcal{O}_X$ be a sheaf of rings on X .

- Define the category of presheaves of $\mathcal{O}_X$ -modules on X to be the category

$$\mathbf{PMod}_{\mathcal{O}_X} = {}_{\mathcal{O}_X}\mathbf{Mod}(\mathbf{PSh}(X))$$

- Define the category of sheaves of $\mathcal{O}_X$ -modules on X to be the category

$$\mathbf{Mod}_{\mathcal{O}_X} = {}_{\mathcal{O}_X}\mathbf{Mod}(\mathbf{Sh}(X))$$

Proposition 1.5.6

Let X be a space. Let $\mathcal{O}_X$ be a sheaf of rings on X . Let $\mathcal{F}$ be presheaf of sets. Then $\mathcal{F}$ is a presheaf of $\mathcal{O}_X$ -modules if and only if the following are true.

- For each open set $U \subseteq X$, we have that $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module.
- For each inclusion of open sets $V \subseteq U$, the morphism $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is a group homomorphism.

Lemma 1.5.7

Let X be a space. Let $\mathcal{O}_X$ be a sheaf of rings on X . Then the following are true.

- Let $\mathcal{F}$ be a presheaf of $\mathcal{O}_X$ -modules. Then $\mathcal{F}_p$ is an $\mathcal{O}_{X,p}$ -module for all $p \in X$.
- Let $\mathcal{F}, \mathcal{G}$ be presheaf of $\mathcal{O}_X$ -modules. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves of $\mathcal{O}_X$ -modules. Then $\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is an $\mathcal{O}_X(U)$ -module homomorphism for all open sets $U \subseteq X$.
- Let $\mathcal{F}, \mathcal{G}$ be presheaf of $\mathcal{O}_X$ -modules. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of presheaves of $\mathcal{O}_X$ -

modules. Then $\varphi_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$ is an $\mathcal{O}_{X,p}$ -module homomorphism for all $p \in X$.

Lemma 1.5.8

Let X be a space. Then the sheafification functor restricts to an adjunction

$$(-)^{\text{sh}} : \mathbf{Mod}_{\mathcal{O}_X} \xrightleftharpoons{\iota} \mathbf{PMod}_{\mathcal{O}_X} : \iota$$

Definition 1.5.9 (The Category of Presheaves and Sheaves of Algebras)

Let X be a space. Let $\mathcal{O}_X$ be a sheaf of commutative rings on X .

- Define the category of presheaves of $\mathcal{O}_X$ -algebras on X to be the category

$$\mathbf{PAlg}_{\mathcal{O}_X} = \mathbf{Alg}_{\mathcal{O}_X}(\mathbf{PSh}(X))$$

- Define the category of sheaves of $\mathcal{O}_X$ -algebras on X to be the category

$$\mathbf{Alg}_{\mathcal{O}_X} = \mathbf{Alg}_{\mathcal{O}_X}(\mathbf{Sh}(X))$$

Proposition 1.5.10

Let X be a space. Then we have $\mathbf{Ring}(X) \cong \mathbf{Alg}_{\mathbb{Z}}$.

Proposition 1.5.11

Let X be a space. Let $\mathcal{O}_X$ be a sheaf of rings on X . Let $\mathcal{A}$ be presheaf of sets. Then $\mathcal{A}$ is a presheaf of $\mathcal{O}_X$ -modules if and only if the following are true.

- For each open set $U \subseteq X$, we have that $\mathcal{A}(U)$ is an $\mathcal{O}_X(U)$ -algebra.
- For each inclusion of open sets $V \subseteq U$, the morphism $\mathcal{A}(U) \rightarrow \mathcal{A}(V)$ is a $\mathcal{O}_X(U)$ -algebra homomorphism.

Lemma 1.5.12

Let X be a space. Let $\mathcal{O}_X$ be a sheaf of rings on X . Then the following are true.

- Let $\mathcal{A}$ be a presheaf of $\mathcal{O}_X$ -algebras. Then $\mathcal{A}_p$ is an $\mathcal{O}_{X,p}$ -algebra for all $p \in X$.
- Let $\mathcal{A}, \mathcal{B}$ be presheaf of $\mathcal{O}_X$ -algebras. Let $\varphi : \mathcal{A} \rightarrow \mathcal{B}$ be a morphism of presheaves of $\mathcal{O}_X$ -algebras. Then $\varphi(U) : \mathcal{A}(U) \rightarrow \mathcal{B}(U)$ is an $\mathcal{O}_X(U)$ -algebra homomorphism for all open sets $U \subseteq X$.
- Let $\mathcal{A}, \mathcal{B}$ be presheaf of $\mathcal{O}_X$ -algebras. Let $\varphi : \mathcal{A} \rightarrow \mathcal{B}$ be a morphism of presheaves of $\mathcal{O}_X$ -algebras. Then $\varphi_p : \mathcal{A}_p \rightarrow \mathcal{B}_p$ is an $\mathcal{O}_{X,p}$ -algebra homomorphism for all $p \in X$.

Lemma 1.5.13

Let X be a space. Then the sheafification functor restricts to an adjunction

$$(-)^{\text{sh}} : \mathbf{Alg}_{\mathcal{O}_X} \xrightleftharpoons{\iota} \mathbf{PAlg}_{\mathcal{O}_X} : \iota$$

2 Sheaves of Modules over a Sheaf of Rings

2.1 The Category of Ringed Spaces

Definition 2.1.1 (Ringed Space)

A ringed space is a topological space X with a sheaf of rings $\mathcal{O}_X$ on X .

Definition 2.1.2 (Morphisms of Ringed Spaces)

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be ringed spaces. A morphism of ringed spaces from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$ is a pair $(f, f^\#)$ of continuous map $f : X \rightarrow Y$ and a map $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ of sheaves of rings on Y .

Definition 2.1.3 (Commutative Ringed Space)

Let $(X, \mathcal{O}_X)$ be a ringed space. We say that X is a commutative ringed space if $\mathcal{O}_X$ is a sheaf of commutative rings.

Definition 2.1.4 (Locally Ringed Spaces)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. We say that X is a locally ringed space if for all $p \in X$, $\mathcal{O}_{X,p}$ is a local ring.

Suppose that $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are locally ringed spaces. Given a point $p \in X$, we obtain induced maps

$$f^\# : \mathcal{O}_Y(V) \rightarrow f_*\mathcal{O}_X(f^{-1}(V))$$

for any neighbourhood V of $f(p)$. Taking the colimit, we obtain a map

$$\mathcal{O}_{Y,f(p)} = \operatorname{colim}_V \mathcal{O}_Y(V) \rightarrow \operatorname{colim}_V \mathcal{O}_X(f^{-1}(V))$$

As V ranges through the neighbourhood of $f(p)$, $f^{-1}(V)$ ranges over a subset of the neighbourhoods of p so that we obtain a map

$$\mathcal{O}_{Y,f(p)} = \operatorname{colim}_V \mathcal{O}_Y(V) \rightarrow \operatorname{colim}_V \mathcal{O}_X(f^{-1}(V)) \rightarrow \mathcal{O}_{X,p}$$

Definition 2.1.5 (Morphisms of Locally Ringed Spaces)

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be locally ringed spaces. Then a morphism of locally ringed spaces is a morphism of ringed spaces such that for each $p \in X$, the induced map of local rings

$$f_p^\# : \mathcal{O}_{Y,f(p)} \rightarrow \mathcal{O}_{X,p}$$

is a local homomorphism of local rings.

Definition 2.1.6 (The Category of Locally Ringed Spaces)

Define the category **LRS** of locally ringed spaces as follows.

- The objects are locally ringed spaces.
- The morphisms are morphisms of locally ringed spaces.
- Composition is given by the composition of continuous functions and composition of natural transformations.

2.2 The Abelian Category Structure of Sheaves of Modules

Definition 2.2.1 (The Presheaf Kernel)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{PMod}_{\mathcal{O}_X}$. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism. Define the kernel $\ker(\varphi)$ of φ as follows.

- For each open set $U \subseteq X$, define $\ker(\varphi)(U) = \ker(\varphi(U))$.
- For each inclusion $V \subseteq U$ of open sets, define the restriction morphism by the universal property of kernels as follows:

$$\begin{array}{ccccc}
\ker(\varphi(U)) & \longrightarrow & \mathcal{F}(U) & \xrightarrow{\varphi(U)} & \mathcal{G}(U) \\
\exists! \downarrow & & \text{res}_V^U \downarrow & & \downarrow \text{res}_V^U \\
\ker(\varphi(V)) & \longrightarrow & \mathcal{F}(V) & \xrightarrow{\varphi(V)} & \mathcal{G}(V)
\end{array}$$

Proposition 2.2.2

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{PMod}_{\mathcal{O}_X}$. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism. Then the following are true.

- $\ker(\varphi)$ is the categorical kernel of φ in $\mathbf{PMod}_{\mathcal{O}_X}$.
- If $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$, then $\ker(\varphi)$ is a sheaf and is the categorical kernel of φ in $\mathbf{Mod}_{\mathcal{O}_X}$.
- $\ker(\varphi)_p = \ker(\varphi_p)$.

Proof

- Let $U \subseteq X$ be an open set. Let $\{U_i \mid i \in I\}$ be an open cover of U . Let $f, g \in \ker(\varphi)(U)$ such that $\text{res}_{U_i}^U(f) = \text{res}_{U_i}^U(g)$. Consider the following commutative diagram:

$$\begin{array}{ccc}
\ker(\varphi(U)) & \longrightarrow & \mathcal{F}(U) \\
\text{res}_V^U \downarrow & & \downarrow \text{res}_V^U \\
\ker(\varphi(U_i)) & \longrightarrow & \mathcal{F}(U_i)
\end{array}$$

Consider the images of f and g in $\mathcal{F}(U)$. Since f and g map vertically down to the same element in $\ker(\varphi)(U_i)$, by commutativity of the diagram, the images of f and g in $\mathcal{F}(U)$ map vertically to the same element in $\mathcal{F}(U_i)$. Since this is true for all $i \in I$, the identity property of the sheaf $\mathcal{F}$ implies that the image of f and g in $\mathcal{F}(U)$ are equal. This means that $f - g \in \ker(\varphi(U))$ map to $0 \in \mathcal{F}(U)$. Since $\ker(\varphi)(U)$ is a subgroup of $\mathcal{F}(U)$, we must have $f = g$. This proves the identity property of $\ker(\varphi)$.

- I claim that the induced map $\ker(\phi)_p \rightarrow \mathcal{F}_p \rightarrow \mathcal{G}_p$ is the zero map. Indeed, the map $\ker(\phi)_p \rightarrow \mathcal{F}_p$ is induced by the maps $\ker(\phi)(U) \rightarrow \mathcal{F}(U)$ for each open set $U \subseteq X$, which composes with $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ to give the zero map. Hence by the universal property of the kernel of ϕ_p , we obtain a unique morphism $\ker(\phi)_p \rightarrow \ker(\phi_p)$. Since the morphism $\ker(\phi) \rightarrow \mathcal{F}$ is the inclusion map for each open set, the morphism $\ker(\phi)_p \rightarrow \mathcal{F}_p$ is also the inclusion map and so $\ker(\phi_p) \subseteq \ker(\phi)_p$. Conversely, suppose that $[(s, U)] \in \ker(\phi_p)$. Then we have $[(\phi(U)(s), U)] = [(0, V)]$ for some $V \subseteq X$ open. This means that there exists $W \subseteq U \cap V$ open such that $\phi(U)(s)|_W = 0|_W$. But we have

$$\begin{aligned}
0|_W &= \phi(U)(s)|_W \\
&= \text{res}_W^U(\phi(U)(s)) \\
&= \phi(W)(\text{res}_W^U(s)) && (\phi \text{ is a morphism}) \\
&= \phi(W)(s|_W)
\end{aligned}$$

and so $s|_W \in \ker(\phi)(W)$. By definition of the kernel of ϕ , we have $[(s|_W, W)] \in \ker(\phi)_p$. But $[(s|_W, W)] = [(s, U)]$ and so we are done. ■

Definition 2.2.3 (The Presheaf Cokernel)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{PMod}_{\mathcal{O}_X}$. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism. Define the cokernel $\text{coker}(\varphi)$ of φ as follows.

- For each open set $U \subseteq X$, define $\text{coker}(\varphi)(U) = \text{coker}(\varphi(U))$.
- For each inclusion $V \subseteq U$ of open sets, define the restriction morphism by the universal

property of cokernels as follows:

$$\begin{array}{ccccc} \mathcal{F}(U) & \xrightarrow{\varphi(U)} & \mathcal{G}(U) & \longrightarrow & \text{coker}(\varphi(U)) \\ \text{res}_V^U \downarrow & & \downarrow \text{res}_V^U & & \downarrow \exists! \\ \mathcal{F}(V) & \xrightarrow{\varphi(V)} & \mathcal{G}(V) & \longrightarrow & \text{coker}(\varphi(V)) \end{array}$$

Proposition 2.2.4

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{PMod}_{\mathcal{O}_X}$. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism. Then the following are true.

- $\text{coker}(\varphi)$ is the categorical cokernel of φ in $\mathbf{PMod}_{\mathcal{O}_X}$.
- If $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$, then $\text{coker}(\varphi)^{\text{sh}}$ is a sheaf and is the categorical cokernel of φ in $\mathbf{Mod}_{\mathcal{O}_X}$.
- If $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$, then $\text{coker}(\varphi)_p^{\text{sh}} = \text{coker}(\varphi_p)$.

Proof

- We prove that $\text{coker}(\varphi)^{\text{sh}}$ satisfy the universal property of the cokernel. Let $\mathcal{H}$ be a sheaf on X with values in $\mathcal{A}$. Let $\psi : \mathcal{G} \rightarrow \mathcal{H}$ be a morphism such that the composite $\mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H}$ is 0. Consider the following diagram:

$$\begin{array}{ccccc} \mathcal{F} & \xrightarrow{\varphi} & \mathcal{G} & \longrightarrow & \text{coker}(\varphi) & \xrightarrow{\theta_{\text{coker}(\varphi)}} & \text{coker}(\varphi)^{\text{sh}} \\ & & \searrow \psi & & \downarrow \exists! & & \downarrow \exists! \\ & & & & & & \mathcal{H} \end{array}$$

By the universal property of cokernels in the category of presheaves, there exists a unique morphism $\text{coker}(\varphi) \rightarrow \mathcal{H}$ such that the relevant diagram commutes. Then by the universal property of sheafification, we obtain the unique morphism required to satisfy the universal property of the cokernel.

- Colimits commute with colimits.

■

Proposition 2.2.5

Let X be a space. Then $\mathbf{Mod}_{\mathcal{O}_X}$ is an abelian category.

Proof We showed that $\mathbf{Mod}_{\mathcal{O}_X}$ has a zero object by 2.2.1. The hom sets have the structure of an abelian group by open set wise, point wise addition. Thus addition distributes over composition. Finite products are given in def 2.2.2 and lmm 2.2.3. Kernels and cokernels are given above for arbitrary morphisms. Now for any monomorphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ of sheaves,

■

Let $\mathcal{A}$ be an abelian category. Let f be a morphism. Recall the following equivalent definitions.

- f is injective if and only if $\ker(f) = 0$ if and only if f is a monomorphism.
- f is surjective if and only if $\text{coker}(f) = 0$ if and only if f is an epimorphism.
- f is an isomorphism if and only if f is injective and surjective.

If $\mathcal{A} = \mathbf{Set}$ recall also that injective functions and monomorphisms coincide, while surjective functions and monomorphisms coincide.

Definition 2.2.6 (The Image Presheaf)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{PMod}_{\mathcal{O}_X}$. Define the image presheaf $\text{im}(\varphi)$ by the following.

- For each open set $U \subseteq X$, define $\text{im}(\varphi)(U) = \text{im}(\varphi(U))$.
- For each inclusion of open sets $V \subseteq U$, define $\text{im}(\varphi)(U) \rightarrow \text{im}(\varphi)(V)$ by the universal

property of kernels as follows:

$$\begin{array}{ccccc} \mathrm{im}(\varphi)(U) = \ker(\mathcal{G}(U) \rightarrow \mathrm{coker}(f)(U)) & \longrightarrow & \mathcal{G}(U) & \longrightarrow & \mathrm{coker}(\varphi)(U) \\ \exists! \downarrow & & \mathrm{res}_V^U \downarrow & & \downarrow \mathrm{res}_V^U \\ \mathrm{im}(\varphi)(V) = \ker(\mathcal{G}(V) \rightarrow \mathrm{coker}(f)(V)) & \longrightarrow & \mathcal{G}(V) & \longrightarrow & \mathrm{coker}(\varphi)(V) \end{array}$$

Proposition 2.2.7

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{PMod}_{\mathcal{O}_X}$. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism. Then the following are true.

- $\mathrm{im}(\varphi)$ is the image of φ in $\mathbf{PMod}_{\mathcal{O}_X}$.
- If $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$, then $\mathrm{im}(\varphi)^{\mathrm{sh}}$ is the image of φ in $\mathbf{Mod}_{\mathcal{O}_X}$.
- There is a natural isomorphism $\mathrm{im}(\varphi)_p^{\mathrm{sh}} = \mathrm{im}(\varphi_p)$.

Proof

-
- Since the kernel of the cokernel is the cokernel of the kernel in an abelian category, we have that

$$\mathrm{im}(\varphi) = \ker(\mathrm{coker}(\varphi)^{\mathrm{sh}}) = \mathrm{coker}(\ker(\varphi))^{\mathrm{sh}} = \ker(\mathrm{coker}(\varphi))^{\mathrm{sh}}$$

- Since taking kernels and cokernels commute with stalk, and sheafification preserves stalks, we conclude.

■

Proposition 2.2.8

Let X be a space. Suppose the following

$$\mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H}$$

is a sequence in $\mathbf{Mod}_{\mathcal{O}_X}$. Then the sequence is exact if and only if the following sequence of stalks

$$\mathcal{F}_p \longrightarrow \mathcal{G}_p \longrightarrow \mathcal{H}_p$$

is exact for all $p \in X$.

Proof We have that $\mathrm{im}(\mathcal{F} \rightarrow \mathcal{G}) = \ker(\mathcal{G} \rightarrow \mathcal{H})$ and taking stalks gives $\mathrm{im}(\mathcal{F}_p \rightarrow \mathcal{G}_p) = \ker(\mathcal{G}_p \rightarrow \mathcal{H}_p)$. Since taking stalks commute with images and kernels, we are done. Conversely, suppose that for all $p \in X$, we have $\mathrm{im}(\mathcal{F}_p \rightarrow \mathcal{G}_p) = \ker(\mathcal{G}_p \rightarrow \mathcal{H}_p)$. Since taking stalks commute with kernels and images, this show that $\mathrm{im}(\mathcal{F} \rightarrow \mathcal{G})_p = \ker(\mathcal{G} \rightarrow \mathcal{H})_p$. But two sheaves are isomorphic if and only if the stalks are isomorphic, we are done. ■

Thus the functor

$$(-)_p : \mathbf{Mod}_{\mathcal{O}_X} \rightarrow \mathbf{Mod}_{\mathcal{O}_{X,p}}$$

of taking stalks is exact.

Proposition 2.2.9

Let X be a space. Let the following be a sequence in $\mathbf{Mod}_{\mathcal{O}_X}$:

$$0 \longrightarrow \mathcal{F} \xrightarrow{\varphi} \mathcal{G} \xrightarrow{\psi} \mathcal{H}$$

Then the above sequence is exact if and only if the following sequence

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{\varphi(U)} \mathcal{G}(U) \xrightarrow{\psi(U)} \mathcal{H}(U)$$

of local sections is exact for any open set $U \subseteq X$.

Proof

Suppose that the given sequence of sheaves is exact. This means that $\ker(\mathcal{F} \rightarrow \mathcal{G}) = 0$. Let $U \subseteq X$ be an open set. Then we have $0 = \ker(\varphi)(U) = \ker(\varphi(U))$. On the other hand, notice that $\text{im}(\varphi) = \ker(\psi)$ implies that $\psi_p \circ \varphi_p = 0$ for all $p \in X$. Recall that the image is sheafified, meaning that an element in the image is

$$([(s_p, U_p)])_{p \in U} \in \prod_{p \in U} \text{im}(\varphi)_p$$

This means that $[(s_p, U_p)] \in \text{im}(\varphi)_p$. So there exists $[(t_p, V_p)] \in \mathcal{F}_p$ such that $\varphi_p([(t_p, V_p)]) = [(s_p, U_p)]$. Also, since the sheafification of a sheaf is isomorphic to the original sheaf, without loss of generality, Then we have

$$\begin{aligned} \psi(U) (\varphi_p([(t_p, V_p)]))_{p \in U} &= \psi(U) ([(\varphi(V_p)(t_p), V_p)])_{p \in U} \\ &= ([(\psi(U)(\varphi(U)(t_p)), V_p)])_{p \in U} \\ &= ([(\psi \circ \varphi(U)(t_p), V_p)])_{p \in U} \\ &= 0 \end{aligned}$$

Thus $\text{im}(\varphi)(U) \subseteq \ker(\psi)(U)$. Let $s \in \ker(\psi)(U)$. ■

Thus the functor

$$(-)(U) : \mathbf{Mod}_{\mathcal{O}_X} \rightarrow \mathbf{Mod}_{\mathcal{O}_X(U)}$$

is left exact. It is in general not right exact.

Proposition 2.2.10

Let X be a topological space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism. Then the following are equivalent.

- φ is injective (a monomorphism).
- For each $U \subseteq X$ open, $\varphi(U)$ is injective.
- For each $p \in X$, φ_p is injective.

Proof (1) $\iff$ (2) is a special case of 2.3.2. (1) $\iff$ (3) is a special case of 2.3.1. ■

Proposition 2.2.11

Let X be a topological space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism. Then the following are equivalent.

- φ is surjective (an epimorphism).
- For each $U \subseteq X$ open and for every $g \in \mathcal{G}(U)$, there exists an open cover $\{U_i \mid i \in I\}$ of U and there exists $f_i \in \mathcal{F}(U_i)$ such that $\varphi(U_i)(f_i) = g|_{U_i}$.
- For each $p \in X$, φ_p is surjective.
- There exists a base $\mathcal{B} = \{B_i \mid i \in I\}$ of X such that $\varphi(B_i)$ is surjective for all $i \in I$.

2.3 Graded Sheaves

Definition 2.3.1 (Graded Sheaves of $\mathcal{O}_X$ -Modules)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. We say that $\mathcal{F}$ is graded if the following are true.

- $\mathcal{F}$ admits a decomposition

$$\mathcal{F} = \bigoplus_{i=-\infty}^{\infty} \mathcal{F}_i$$

where each $\mathcal{F}_i$ is a sheaf of $\mathcal{O}_X$ -modules.

- For each open set $U \subseteq X$, we have that $\mathcal{F}(U)$ admits a grading

$$\mathcal{F}(U) = \bigoplus_{i=-\infty}^{\infty} \mathcal{F}_i(U)$$

as a $\mathcal{O}_X(U)$ -module.

- For each inclusion $V \subseteq U$ of open sets in X , the induced map $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ restricts to maps $\mathcal{F}_i(U) \rightarrow \mathcal{F}_i(V)$ for all i .

Definition 2.3.2 (Graded Sheaves of $\mathcal{O}_X$ -Algebras)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -algebras. We say that $\mathcal{F}$ is graded if the following are true.

- $\mathcal{F}$ is graded as a $\mathcal{O}_X$ -module.
- The map $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{F} \rightarrow \mathcal{F}$ restricts to maps $\mathcal{F}_i \otimes_{\mathcal{O}_X} \mathcal{F}_j \rightarrow \mathcal{F}_{i+j}$.

2.4 Tensor-Hom Adjunction

Definition 2.4.1 (Tensor Products)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$. Define the tensor product $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ of $\mathcal{F}$ and $\mathcal{G}$ to be the sheafification of the sheaf defined as follows.

- For each open set $U \subseteq X$, define $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})(U) = \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$.
- For each inclusion of open sets $V \subseteq U$, define the morphism $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})(U) \rightarrow (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})(V)$ by

Tensor products commute with stalks.

Definition 2.4.2 (The Sheaf Hom)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$. Define the sheaf hom from $\mathcal{F}$ to $\mathcal{G}$ to be the sheaf

$$\mathcal{H}om(\mathcal{F}, \mathcal{G})$$

defined as follows.

- Each open set U is sent to the abelian group

$$\mathcal{H}om(\mathcal{F}, \mathcal{G})(U) = \mathcal{H}om_{\mathbf{Mod}_{\mathcal{O}_X|_U}}(\mathcal{F}|_U, \mathcal{G}|_U)$$

- For each inclusion of open sets $V \hookrightarrow U$, define the morphism

$$\mathcal{H}om_{\mathbf{Mod}_{\mathcal{O}_X}}(\mathcal{F}|_U, \mathcal{G}|_U) \rightarrow \mathcal{H}om_{\mathbf{Mod}_{\mathcal{O}_X}}(\mathcal{F}, \mathcal{G})(V)$$

as follows. A morphism of sheaves $\phi : \mathcal{F}|_U \rightarrow \mathcal{G}|_U$ is sent to the restriction $\phi|_V : \mathcal{F}|_V \rightarrow \mathcal{G}|_V$.

Definition 2.4.3 (The Dual Sheaf)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Define the dual sheaf of $\mathcal{F}$ to be

$$\mathcal{F}^\vee = \mathcal{H}om(\mathcal{F}, \mathcal{O}_X)$$

Theorem 2.4.4 (The Tensor-Hom Adjunction)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G}, \mathcal{H} \in \mathbf{Mod}_{\mathcal{O}_X}$. Then there is a natural isomorphism

$$\mathbf{Hom}_{\mathbf{Mod}_{\mathcal{O}_X}}(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}, \mathcal{H}) \cong \mathbf{Hom}_{\mathbf{Mod}_{\mathcal{O}_X}}(\mathcal{F}, \mathcal{H}om_{\mathbf{Mod}_{\mathcal{O}_X}}(\mathcal{G}, \mathcal{H}))$$

Proposition 2.4.5

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{A}, \mathcal{B} \in \mathbf{Alg}_{\mathcal{O}_X}$. Then $\mathcal{H}om_{\mathbf{Alg}_{\mathcal{O}_X}}(\mathcal{A}, \mathcal{B}) \in$

$\mathbf{Alg}_{\mathcal{O}_X}$.

Proposition 2.4.6

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathbf{Alg}_{\mathcal{O}_X}$. Then we have

$$\mathcal{H}om_{\mathbf{Alg}_{\mathcal{O}_X}}(\mathcal{A} \otimes_{\mathcal{O}_X} \mathcal{B}, \mathcal{C}) \cong \mathcal{H}om_{\mathbf{Alg}_{\mathcal{O}_X}}(\mathcal{A}, \mathcal{C}) \times \mathcal{H}om_{\mathbf{Alg}_{\mathcal{O}_X}}(\mathcal{B}, \mathcal{C})$$

2.5 Pushforward and Pullback Sheaves of Modules

Lemma 2.5.1

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Then inverse and direct image adjunction restricts to an adjunction

$$f^{-1} : \mathbf{Ab}(Y) \overset{\perp}{\rightleftarrows} \mathbf{Ab}(X) : f_*$$

However for modules in general, we need to modify the inverse image functor to incorporate the action of the ring.

Definition 2.5.2 (Inverse Image Sheaf)

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be commutative ringed spaces. Let $f : X \rightarrow Y$ be a morphism. Let $\mathcal{G}$ be a sheaf on Y . Define the inverse image sheaf of $\mathcal{G}$ by

$$f^*\mathcal{G} = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{G}$$

Since tensor products are functorial, the inverse image sheaf is also a functorial construction. Also, this is basically a change of rings formula. In particular, the underlying sheaf of abelian groups of $f^{-1}\mathcal{G}$ and $f^*\mathcal{G}$ are the same.

Lemma 2.5.3

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Then there is an adjunction

$$f^* : \mathbf{Mod}_{\mathcal{O}_Y} \overset{\perp}{\rightleftarrows} \mathbf{Mod}_{\mathcal{O}_X} : f_*$$

Lemma 2.5.4

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Then inverse and direct image adjunction restricts to an adjunction

$$f^* : \mathbf{Alg}_{\mathcal{O}_Y} \overset{\perp}{\rightleftarrows} \mathbf{Alg}_{\mathcal{O}_X} : f_*$$

2.6 Tensor Algebra and its Subalgebras

From operad theory a lot of constructions are carried over eg symmetric products, exterior products.

Definition 2.6.1 (k th Tensor Power)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Define the k th tensor power of $\mathcal{F}$ to be sheafification of the presheaf

$$U \mapsto T^k(\mathcal{F}(U))$$

Lemma 2.6.2

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. If $\mathcal{F}$ is locally free of rank n , then $T^k(\mathcal{F})$ is locally free of rank n^k .

Definition 2.6.3 (Tensor Algebra)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Define the tensor

algebra of $\mathcal{F}$ to be sheafification of the presheaf

$$U \mapsto T(\mathcal{F}(U))$$

Definition 2.6.4 (*k*th Symmetric Power)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Define the *k*th symmetric power of $\mathcal{F}$ to be sheafification of the presheaf

$$U \mapsto S^k(\mathcal{F}(U))$$

Lemma 2.6.5

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. If $\mathcal{F}$ is locally free of rank n , then $T^k(\mathcal{F})$ is locally free of rank $\binom{n+k-1}{n-1}$.

Definition 2.6.6 (Symmetric Algebra)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Define the symmetric algebra of $\mathcal{F}$ to be sheafification of the presheaf

$$U \mapsto S(\mathcal{F}(U))$$

Definition 2.6.7 (*k*th Exterior Power)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Define the *k*th exterior power of $\mathcal{F}$ to be sheafification of the presheaf

$$U \mapsto \Lambda^k(\mathcal{F}(U))$$

Lemma 2.6.8

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. If $\mathcal{F}$ is locally free of rank n , then the following are true.

- $\Lambda^k(\mathcal{F})$ is locally free of rank $\binom{n}{k}$.
- The multiplication $\Lambda^r(\mathcal{F}) \otimes_{\mathcal{O}_X} \Lambda^{n-r}(\mathcal{F}) \rightarrow \Lambda^n(\mathcal{F})$ is an isomorphism for any r .

Definition 2.6.9 (Exterior Algebra)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Define the exterior algebra of $\mathcal{F}$ to be sheafification of the presheaf

$$U \mapsto \Lambda(\mathcal{F}(U))$$

all the above commutes with pullbacks of commutative ringed spaces.

3 Different Types of Sheaves of Modules

3.1 Some Finiteness Properties

Definition 3.1.1 (Sheaf of Finite Type)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. We say that $\mathcal{F}$ is a sheaf of finite type if there exists an open cover $\{U_i \mid i \in I\}$ of X such that there is a surjective morphism

$$\mathcal{O}_X^{\otimes n_i}|_{U_i} \rightarrow \mathcal{F}|_{U_i}$$

for some $n_i \in \mathbb{N}$.

Proposition 3.1.2

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$ be a sheaf of finite type. Then $\mathcal{F}_{X,p}$ is a finitely generated $\mathcal{O}_{X,p}$ -module for all $p \in X$.

Definition 3.1.3 (Finitely Presented Sheaves)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. We say that $\mathcal{F}$ is finitely presented if there exists an open cover $\{U_i \mid i \in I\}$ of X such that there is an exact sequence

$$\mathcal{O}_X^{m_i}|_U \longrightarrow \mathcal{O}_X^{n_i}|_U \longrightarrow \mathcal{F}|_U \longrightarrow 0$$

for some $m_i, n_i \in \mathbb{N}$.

3.2 Quasi-Coherent Sheaves

Quasi-coherent sheaves and coherent sheaves has extremely abstract definitions. If one is only interested in the case of Algebraic Geometry, then there is a very nice explicit way of defining it, so readers may skip this part.

Definition 3.2.1 (Quasi-Coherent Sheaves)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -module. We say that $\mathcal{F}$ is a quasi-coherent sheaf if for all $p \in X$, there exists an open neighbourhood $U \subseteq X$ such that there is an exact sequence:

$$\mathcal{O}_X^{\otimes I}|_U \longrightarrow \mathcal{O}_X^{\otimes J}|_U \longrightarrow \mathcal{F}|_U \longrightarrow 0$$

for some countable indexing sets I and J .

Definition 3.2.2 (The Category of Quasi-coherent Sheaves)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Define the category of coherent sheaves $\mathbf{QCoh}_{\mathcal{O}_X}$ to be the full subcategory of $\mathbf{Mod}_{\mathcal{O}_X}$ consisting of quasi-coherent sheaves.

Definition 3.2.3 (Coherent Sheaves)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -module. We say that $\mathcal{F}$ is coherent if the following are true.

- $\mathcal{F}$ is a sheaf of finite type.
- For any $U \subseteq X$ and any morphism

$$\varphi : \mathcal{O}_X^{\otimes n}|_U \rightarrow \mathcal{F}|_U$$

of $\mathcal{O}_X$ -modules, then kernel of φ is a sheaf of finite type.

Definition 3.2.4 (The Category of Coherent Sheaves)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Define the category of coherent sheaves $\mathbf{Coh}_{\mathcal{O}_X}$ to be the full subcategory of $\mathbf{QCoh}_{\mathcal{O}_X}$ consisting of coherent sheaves.

Proposition 3.2.5

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let the following

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F}'' \longrightarrow 0$$

be an exact sequence of sheaves. If two of the three sheaves are coherent, then so is the last one.

Proposition 3.2.6

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. Then the following are true.

- If $\mathcal{F}$ is coherent, then $\mathcal{F}$ is finitely presented.
- If $\mathcal{F}$ is finitely presented, then $\mathcal{F}$ is of finite type.
- If $\mathcal{F}$ is finitely presented, then $\mathcal{F}$ is quasi-coherent.

Proposition 3.2.7

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. If $\mathcal{O}_X$ is coherent, then the following are equivalent.

- $\mathcal{F}$ is coherent.
- $\mathcal{F}$ is finitely presented.
- $\mathcal{F}$ is of finite type.

3.3 Reflexive Sheaves

Definition 3.3.1 (Reflexive Sheaves)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. We say that $\mathcal{F}$ is reflexive if there is an isomorphism of $\mathcal{O}_X$ -modules

$$\mathcal{F}^{\vee\vee} \cong \mathcal{F}$$

3.4 Locally Free Sheaves

Definition 3.4.1 (Free Sheaves)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a $\mathcal{O}_X$ -module. We say that $\mathcal{F}$ is free if there is an isomorphism of $\mathcal{O}_X$ -modules

$$\mathcal{F} \cong \mathcal{O}_X^{\oplus n}$$

Definition 3.4.2 (Locally Free Sheaves)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. We say that $\mathcal{F}$ is locally free if X has an open cover $\{U_i \mid i \in I\}$ such that for each U_i , there is an isomorphism of $\mathcal{O}_X$ -modules

$$\mathcal{F}|_{U_i} \cong (\mathcal{O}_X|_{U_i})^{\oplus n_i}$$

Lemma 3.4.3

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a locally free sheaf. Then $\mathcal{F}$ is reflexive.

Definition 3.4.4 (The Rank of a Locally Free Sheaf)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a locally free sheaf of $\mathcal{O}_X$ -modules. We say that the rank of $\mathcal{F}$ is $n \in \mathbb{N}$ if there exists an open cover $\{U_i \mid i \in I\}$ of X such that there are isomorphisms of $\mathcal{O}_X$ -modules

$$\mathcal{F}|_{U_i} \cong (\mathcal{O}_X|_{U_i})^{\oplus n}$$

Lemma 3.4.5

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a locally free sheaf of $\mathcal{O}_X$ -modules. If X is connected, then the rank of $\mathcal{F}$ is well defined.

Proposition 3.4.6 Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Then $\mathcal{F}$ is locally free of rank r if and only if

$$\mathcal{F}_p \cong \mathcal{O}_{X,p}^r$$

for all $p \in X$.

Proposition 3.4.7

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. If $\mathcal{F}$ is locally free, then $\mathcal{F}$ is reflexive.

Definition 3.4.8 (The Category of Locally Free Sheaves)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Define the category of locally free sheaves

$$\mathbf{Loc}_X$$

of rank n over X as follows.

- The objects are the locally free sheaves over X
- The morphisms are the morphisms of sheaves of modules
- Composition is given by the composition of morphisms of sheaves.

3.5 Invertible Sheaves

Definition 3.5.1 (Invertible Sheaf)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. We say that $\mathcal{F}$ is an invertible sheaf if $\mathcal{F}$ is a locally free sheaf of rank 1.

Proposition 3.5.2

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Then the following are equivalent characterization of a sheaf of $\mathcal{O}_X$ -modules $\mathcal{F}$

- $\mathcal{F}$ is invertible
- There exists a sheaf G such that $\mathcal{F} \otimes_{\mathcal{O}_X} G \cong \mathcal{O}_X$
- $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{F}^\vee \cong \mathcal{O}_X$

Proposition 3.5.3

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}$ and $\mathcal{G}$ be invertible sheaves on X . Then $\mathcal{F} \otimes \mathcal{G}$ is an invertible sheaf on X .

Definition 3.5.4 (The Picard Group) Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Define the Picard group of X to be the set

$$\mathrm{Pic}(X) = \{\mathcal{L} \mid \mathcal{L} \text{ is an invertible sheaf on } X\}$$

together with the tensor product $\otimes$.

4 Sheaf Cohomology

4.1 Basic Definitions

Proposition 4.1.1

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Then $\mathbf{Mod}_{\mathcal{O}_X}$ has enough injectives.

Definition 4.1.2 (Global Section Functor)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Define the global section functor $\Gamma_{\mathcal{O}_X}(X, -) : \mathbf{Mod}_{\mathcal{O}_X} \rightarrow \mathbf{Mod}_{\mathcal{O}_X(X)}$ by the following data.

- For $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$, we have $\Gamma_{\mathcal{O}_X}(X, \mathcal{F}) = \mathcal{F}(X)$.
- For a morphism of sheaves $\phi : \mathcal{F} \rightarrow \mathcal{G}$, $\Gamma_{\mathcal{O}_X}(X, -)$ sends it to the group homomorphism $\phi(X) : \mathcal{F}(X) \rightarrow \mathcal{G}(X)$

In general, we can define functor by sending every sheaf to a the local section of a chosen open set. This operation in general is left exact. This means that we have the following proposition.

Corollary 4.1.3

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. The global section functor $\Gamma_{\mathcal{O}_X}(X, -) : \mathbf{Mod}_{\mathcal{O}_X} \rightarrow \mathbf{Mod}_{\mathcal{O}_X(X)}$ is a left exact functor.

Proof The global section functor is a special case of the above proposition. ■

Definition 4.1.4 (Cohomology Functors)

Let X be a space. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X(X)}$. Define the sheaf cohomology of $\mathcal{F}$ to be

$$H^i(X, \mathcal{F}) = R^i \Gamma_{\mathcal{O}_X}(X, -)(\mathcal{F})$$

4.2 Flasque Sheaves

Definition 4.2.1 (Flasque Sheaves)

Let X be a space. Let $\mathcal{F} \in \mathbf{Ab}(X)$. We say that $\mathcal{F}$ is flasque if for any open sets $V \subseteq U \subseteq X$, we have that $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is surjective.

Lemma 4.2.2

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. Then $\mathcal{F}$ is flasque.

Lemma 4.2.3

Let X be a space and let $\mathcal{F}, \mathcal{F}', \mathcal{F}''$ be sheaves on X such that there is an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F}'' \longrightarrow 0$$

of sheaves. Then the following are true.

- If $\mathcal{F}$ and $\mathcal{F}'$ are flasque, then $\mathcal{F}''$ is flasque.
- If $\mathcal{F}$ is flasque, then for any $U \subseteq X$ open, there is an exact sequence

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{\phi} \mathcal{F}'(U) \xrightarrow{\psi} \mathcal{F}''(U) \longrightarrow 0$$

Proof

Suppose that $\mathcal{F}$ and $\mathcal{F}'$ are flasque. Then by naturality we obtain a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{F}'(U) & \xrightarrow{\psi(U)} & \mathcal{F}''(U) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{F}(V) & \xrightarrow{\phi(V)} & \mathcal{F}'(V) & \xrightarrow{\psi(V)} & \mathcal{F}''(V) \longrightarrow 0 \end{array}$$

for any $V \subseteq U$ open sets. Let $s \in \mathcal{F}''(V)$. Then by surjectivity of $\psi(V)$ and $\mathcal{F}'(U) \rightarrow \mathcal{F}'(V)$, there exists $t \in \mathcal{F}'(U)$ that maps correspondingly by the same maps to s . Then using the map $\phi(U)$ we

can send s to an element $r \in \mathcal{F}''(U)$ such that r is sent to s under the map $\mathcal{F}''(U) \rightarrow \mathcal{F}''(V)$. Thus this map is surjective and $\mathcal{F}''$ is flasque.

Now suppose that $\mathcal{F}$ is flasque. ■

Proposition 4.2.4

Let X be a space and let $\mathcal{F}$ be a sheaf on X . If $\mathcal{F}$ is flasque then $\mathcal{F}$ is acyclic for $\Gamma(X, -)$. Explicitly, this means that

$$H^i(X, \mathcal{F}) = 0$$

for all $i > 0$.

Proof

Since $\mathbf{Ab}(X)$ has enough injective objects, suppose that $\mathcal{I}$ is an injective object of $\mathbf{Ab}(X)$ such that $\mathcal{F}$ is a subobject of $\mathcal{I}$. Let $\mathcal{G}$ be the quotient sheaf $\mathcal{F}/\mathcal{I}$. Then there is an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I} \longrightarrow \mathcal{G} \longrightarrow 0$$

By lemma 3.2.2, $\mathcal{I}$ is flasque. By 3.2.3, $\mathcal{G}$ is flasque. By 3.2.3, there is an exact sequence

$$0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{I}(X) \longrightarrow \mathcal{G}(X) \longrightarrow 0$$

Since $\mathcal{I}$ is injective, we have that $H^i(X, \mathcal{I}) = 0$ for all $i > 0$. By passing to the long exact sequence in cohomology, we deduce that $H^1(X, \mathcal{F}) = 0$ and $H^i(X, \mathcal{F}) = H^{i-1}(X, \mathcal{G})$ for each $i \geq 2$. Applying the same method again gives $H^1(X, \mathcal{G}) = 0$. Thus by induction, we conclude that $H^i(X, \mathcal{F}) = 0$ for all $i > 0$. ■

Acyclic sheaves are useful because they help compute sheaf cohomology. Recall from Homological Algebra that

$$H^i(X; \mathcal{F}) = H^i(\Gamma(X, \mathcal{I}^\bullet))$$

if $\mathcal{I}^\bullet$ is an acyclic resolution of $\mathcal{F}$.

4.3 Godement Resolution

Definition 4.3.1 (The Godement Sheaf)

Let X be a space. Let $\mathcal{F} \in \mathbf{Ab}(X)$ be a sheaf on X . Define the Godement sheaf $\mathrm{Gd}(\mathcal{F}) : \mathbf{Open}(X) \rightarrow \mathbf{Ab}$ as follows.

- For each open set $U \subseteq X$, define

$$\mathrm{Gd}(\mathcal{F})(U) = \prod_{x \in U} \mathcal{F}_x$$

- For each inclusion $V \subseteq U$, define the induced map $\mathrm{Gd}(\mathcal{F})(U) \rightarrow \mathrm{Gd}(\mathcal{F})(V)$ to be the restriction map.

Lemma 4.3.2

Let X be a space. Let $\mathcal{F} \in \mathbf{Ab}(X)$ be a sheaf on X . Then $\mathrm{Gd}(\mathcal{F})$ is a flasque sheaf.

Definition 4.3.3 (Canonical Morphism)

Let X be a space. Let $\mathcal{F} \in \mathbf{Ab}(X)$ be a sheaf on X . Define the canonical morphism $\alpha_{\mathcal{F}} : \mathcal{F} \rightarrow \mathrm{Gd}(\mathcal{F})$ by $\alpha_{\mathcal{F}} : \mathcal{F}(U) \rightarrow \mathrm{Gd}(\mathcal{F})(U)$ the formula

$$s \mapsto (s_x)_{x \in U}$$

Definition 4.3.4 (The Godement Functor)

Let X be a space. Define the Godement functor $\mathrm{Gd} : \mathbf{Ab}(X) \rightarrow \mathbf{Ab}(X)$ as follows.

- Each sheaf $\mathcal{F} \in \mathbf{Ab}(X)$ is sent to the godement sheaf $\mathrm{Gd}(\mathcal{F})$.
- Each morphism $\mathcal{F} \rightarrow \mathcal{G}$ is sent to the morphism of sheaves defined by the product of the induced map on stalks.

Proposition 4.3.5

Let X be a space. Let $\mathcal{F} \in \mathbf{Ab}(X)$ be a sheaf. Then $\mathrm{Gd}(\mathcal{F})$ is flasque.

Definition 4.3.6 (Godement Resolution)

Let X be a space. Let $\mathcal{F} \in \mathbf{Ab}(X)$ be a sheaf. Define the Godement resolution $\mathrm{Gd}(\mathcal{F})^\bullet$ of $\mathcal{F}$ as follows.

- Define $\mathrm{Gd}(\mathcal{F})^0 = \mathcal{F}$.
- For each $n \in \mathbb{N}^+$, define $\mathrm{Gd}(\mathcal{F})^n = \mathrm{Gd}(\mathrm{coker}(\alpha_{\mathrm{Gd}(\mathcal{F})^{n-1}}))$.
- Define the morphisms $\mathrm{Gd}(\mathcal{F})^n \rightarrow \mathrm{Gd}(\mathcal{F})^{n+1}$ to be $\alpha_{\mathrm{Gd}(\mathcal{F})^n}$.

Corollary 4.3.7

Let X be a space. Let $\mathcal{F} \in \mathbf{Ab}(X)$ be a sheaf. Then $\mathrm{Gd}(\mathcal{F})^\bullet$ is an acyclic resolution for $\Gamma(X, -)$.

Consequently, the sheaf cohomology of $\mathcal{F}$ acquires the formula

$$H^i(X; \mathcal{F}) = H^i(\Gamma(X, \mathrm{Gd}(\mathcal{F})^\bullet))$$

4.4 Čech Cohomology

Definition 4.4.1 (Čech Complex)

Let X be a space. Let $\mathcal{U} = \{U_i \mid i \in I\}$ be an open cover of X where I is an indexing set. Let $\mathcal{F} \in \mathbf{Ab}(X)$. Define the Čech complex

$$(C^\bullet(X, \mathcal{U}, \mathcal{F}), d) \in \mathbf{Ch}(\mathbf{Ab})$$

of $\mathcal{F}$ with respect to $\mathcal{U}$ as follows.

- The terms of the chain complex is given by

$$C^k(X, \mathcal{U}, \mathcal{F}) = \prod_{i_0 < \dots < i_k} \mathcal{F}(U_{i_0} \cap U_{i_1} \cap \dots \cap U_{i_k})$$

- Define the boundary map $d : C^k(X, \mathcal{U}, \mathcal{F}) \rightarrow C^{k+1}(X, \mathcal{U}, \mathcal{F})$ as follows. For $s \in C^k(X, \mathcal{U}, \mathcal{F})$ whose components are indicated by $s_{i_0, \dots, i_k}$, the components of $d(s)$ is given by

$$(d(s))_{i_0, \dots, i_{k+1}} = \sum_{j=0}^{k+1} (-1)^j s_{i_0, \dots, i_{j-1}, \widehat{i_j}, i_{j+1}, \dots, i_{k+1}}|_{U_{i_0}, \dots, i_{k+1}}$$

We use the notation $U_{i_0, \dots, i_k} = U_{i_0} \cap U_{i_1} \cap \dots \cap U_{i_k}$.

Lemma 4.4.2 For any space X and any open cover $\mathcal{U}$ of X , $(C^\bullet(X, \mathcal{U}, \mathcal{F}), d)$ is indeed a chain complex.

Definition 4.4.3 (Čech Cohomology) Let $(C^\bullet(X, \mathcal{U}, \mathcal{F}), d)$ be a Čech complex. Define the k th cohomology group of it to be

$$\check{H}^k(X, \mathcal{U}, \mathcal{F}) = \frac{\ker(C^k(X, \mathcal{U}, \mathcal{F}) \rightarrow C^{k+1}(X, \mathcal{U}, \mathcal{F}))}{\mathrm{im}(C^{k-1}(X, \mathcal{U}, \mathcal{F}) \rightarrow C^k(X, \mathcal{U}, \mathcal{F}))} = H(C^\bullet(X, \mathcal{U}, \mathcal{F}), d)$$

Lemma 4.4.4

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{U}$ be an open cover of X . Then we have $\check{H}^0(X, \mathcal{U}, \mathcal{F}) = \mathcal{F}(X)$.

Definition 4.4.5 (The Sheafified Čech Complex)

Let X be a space. Let $\mathcal{U}$ be an open cover of X . Let $\mathcal{F}$ be a sheaf on X . Define the sheafified Čech

complex $\mathcal{C}^\bullet(X, \mathcal{U}, \mathcal{F})$ by the following.

- For each $p \in \mathbb{N}$, define

$$\mathcal{C}^p(X, \mathcal{U}, \mathcal{F}) = \prod_{i_0 < \dots < i_p} \iota_{U_{i_0, \dots, i_p}}(U_{i_0, \dots, i_p}) \in \mathbf{Ab}(X)$$

where $\iota_U : U \hookrightarrow X$ denotes the inclusion map for an open subset $U \subseteq X$.

- Define the boundary map $d : \mathcal{C}^p(X, \mathcal{U}, \mathcal{F}) \rightarrow \mathcal{C}^{p+1}(X, \mathcal{U}, \mathcal{F})$ to be the sheafified version of the boundary map in the Čech complex.

Note that by construction, we have $\Gamma(X, \mathcal{C}^p(X, \mathcal{U}, \mathcal{F})) = \mathcal{C}^p(X, \mathcal{U}, \mathcal{F})$.

Lemma 4.4.6

Let X be a space. Let $\mathcal{U}$ be an open cover of X . Let $\mathcal{F}$ be a sheaf on X . Then the following are true.

- $\mathcal{C}^\bullet(X, \mathcal{U}, \mathcal{F})$ is a resolution of $\mathcal{F}$.
- If $\mathcal{F}$ is flasque, then $\mathcal{C}^\bullet(X, \mathcal{U}, \mathcal{F})$ is a flasque (and hence acyclic) resolution of $\mathcal{F}$.

Note that the above lemma implies that $\check{H}^i(X, \mathcal{U}, \mathcal{F}) = H^i(X, \mathcal{F}) = 0$ if $\mathcal{F}$ is flasque.

Lemma 4.4.7

Let X be a space. Let $\mathcal{U}$ be an open cover of X . Let $\mathcal{F}$ be a sheaf on X . Then there exists a map

$$\check{H}^p(X, \mathcal{U}, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$$

that is natural in $\mathcal{F}$ for all $p \in \mathbb{N}$.

Lemma 4.4.8 (Leray)

Let X be a space. Let $\mathcal{U}$ be an open cover of X . Let $\mathcal{F}$ be a sheaf on X . Suppose that for any $U_{i_0, \dots, i_p}$, we have $H^k(U_{i_0, \dots, i_p}, \mathcal{F}_{U_{i_0, \dots, i_p}}) = 0$ for all $k, p \in \mathbb{N}$. Then the above natural maps are isomorphisms

$$\check{H}^p(X, \mathcal{U}, \mathcal{F}) \cong H^p(X, \mathcal{F})$$

Proposition 4.4.9

Let X be a space. Let $\mathcal{F}$ be a sheaf on X . Then there is a natural isomorphism

$$\varinjlim_{\mathcal{U}} \check{H}^1(X, \mathcal{U}, \mathcal{F}) \cong H^1(X, \mathcal{F})$$

Proposition 4.4.10

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Then there is a natural isomorphism

$$\mathrm{Pic}(X) \cong H^1(X, \mathcal{O}_X^*)$$

4.5 Relation to Singular Cohomology

Theorem 4.5.1 ((Y. Sella 2016))

Let X be a locally contractible space. Then there is an isomorphism

$$H_{\mathrm{sing}}^n(X; \mathbb{Z}) \cong H_{\mathrm{sheaf}}^n(X; \mathbb{Z})$$

4.6 Hypercohomology

Definition 4.6.1 (Hypercohomology of a Space)

Let X be a space. Let $\mathcal{F}^\bullet$ be a cochain complex of sheaves on X . Define the hyper cohomology of X to be the right hyper-derived functor of the global section functor:

$$\mathbb{H}^n(X; \mathcal{F}^\bullet) = \mathbb{R}^n \Gamma(\mathcal{F}^\bullet)$$

Proposition 4.6.2

Let X be a space. Let $\mathcal{F}^\bullet \in \mathbf{CCh}^+(\mathbf{Ab}(X))$ be a bounded below cochain complex. Let $\mathcal{I}^\bullet \in \mathbf{CCh}^+(\mathbf{Ab}(X))$ be a bounded below cochain complex such that the following are true.

- $\mathcal{I}^n$ is acyclic for all n .
- There is a quasi-isomorphism $\mathcal{F}^\bullet \rightarrow \mathcal{I}^\bullet$.

Then we have

$$\mathbb{H}^n(X; \mathcal{F}^\bullet) = H^i(\Gamma(X, \mathcal{I}^\bullet))$$

Proposition 4.6.3

Let X be a space. Let $\mathcal{F}^\bullet \in \mathbf{CCh}^+(\mathbf{Ab}(X))$ be a cochain complex of sheaves on X . Then there is a natural isomorphism

$$\mathbb{H}^n(X; \mathcal{F}^\bullet) \cong H^n(\Gamma(X, \mathrm{Tot}^\Pi(\mathrm{Gd}^\bullet, \mathcal{F}^\bullet)))$$

5 More Homological Tools

5.1 Derived Direct Image and Inverse Image

Proposition 5.1.1

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be commutative ringed spaces. Let $f : X \rightarrow Y$ be a morphism. Then f_* is left exact.

Proposition 5.1.2

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be commutative ringed spaces. Let $f : X \rightarrow Y$ be a morphism. Then f^{-1} is exact.

5.2 Ext for Sheaves of Modules

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. Then we have a functor $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, -) : \mathbf{Mod}_{\mathcal{O}_X} \rightarrow \mathbf{Mod}_{\mathcal{O}_X}$ between abelian categories.

Definition 5.2.1 (Ext Functor for Sheaves)

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$. Define the i th ext sheaf of $\mathcal{F}$ and $\mathcal{G}$ to be

$$\mathcal{E}xt_{\mathcal{O}_X}^i(\mathcal{F}, \mathcal{G}) = R^i \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, -)(\mathcal{G})$$

Lemma 5.2.2

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{I} \in \mathbf{Mod}_{\mathcal{O}_X}$ be an injective sheaf. Let $U \subseteq X$ be an open subset. Then $\mathcal{I}|_U$ is injective in $\mathbf{Mod}_{\mathcal{O}_X|_U}$.

Proposition 5.2.3

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$. Let $U \subseteq X$ be open. Then we have

$$\mathcal{E}xt_{\mathcal{O}_X}^i(\mathcal{F}, \mathcal{G})|_{\mathcal{O}_X|_U} \cong \mathcal{E}xt_U^i(\mathcal{F}|_U, \mathcal{G}|_U)$$

Proposition 5.2.4

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. Then we have

$$\mathcal{E}xt_{\mathcal{O}_X}^i(\mathcal{O}_X, \mathcal{F}) = \begin{cases} \mathcal{F} & \text{if } i = 0 \\ 0 & \text{otherwise} \end{cases}$$

Proposition 5.2.5

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F} \in \mathbf{Mod}_{\mathcal{O}_X}$. Then we have

$$\mathrm{Ext}_{\mathcal{O}_X}^i(\mathcal{O}_X, \mathcal{F}) \cong H^i(X, \mathcal{F})$$

Proposition 5.2.6

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G} \in \mathbf{Mod}_{\mathcal{O}_X}$. Let $\mathcal{L}_\bullet$ be a resolution of $\mathcal{F}$ such that each $\mathcal{L}_i$ is locally free of finite rank. Then we have

$$\mathcal{E}xt_{\mathcal{O}_X}^i(\mathcal{F}, \mathcal{G}) \cong H^i(\mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}_\bullet, \mathcal{G}))$$

Proposition 5.2.7

Let $(X, \mathcal{O}_X)$ be a commutative ringed space. Let $\mathcal{F}, \mathcal{G}, \mathcal{L} \in \mathbf{Mod}_{\mathcal{O}_X}$. Suppose that $\mathcal{L}$ is locally free of finite rank. Then the following are true.

- There is an isomorphism

$$\mathrm{Ext}_{\mathcal{O}_X}^i(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}, \mathcal{G}) \cong \mathrm{Ext}_{\mathcal{O}_X}^i(\mathcal{F}, \mathcal{L}^\vee \otimes_{\mathcal{O}_X} \mathcal{G})$$

- There is an isomorphism

$$\mathcal{E}xt_{\mathcal{O}_X}^i(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}, \mathcal{G}) \cong \mathcal{E}xt_{\mathcal{O}_X}^i(\mathcal{F}, \mathcal{L}^\vee \otimes_{\mathcal{O}_X} \mathcal{G}) \cong \mathcal{E}xt_{\mathcal{O}_X}^i(\mathcal{F}, \mathcal{G}) \otimes_{\mathcal{O}_X} \mathcal{L}^\vee$$